

**Poisoned Giants: A Comparative Study of Industrial Pollutants'
Impacts on St Lawrence Estuary Cetaceans**

Introduction

Beluga Whales are often associated with the Arctic region, and they do predominantly inhabit cooler, coastal waters. However, there are a substantial number that live in the St. Lawrence Estuary that exhibit an unusually high rate of intestinal cancer, one of the most contaminated marine populations in the world. Over the course of multiple decades, researchers have found 27% of dead Beluga Whales from the region to have traces of cancer, a rate that is comparable to humans and greater than any other cetacean population in the world (Martineau et al., 2002). These observations have illustrated how even apex predators are threatened at a large-scale by human activities; in this case, environmental contaminants along the shoreline have been found as the cause for inducing the cancer (De Guise et al., 1995).

Historically, factories and other industrial activity local to the St. Lawrence Estuary have discharged waste into the river which includes organochlorines (chemicals that accumulate in animals and can disrupt their hormones/immunity), PAHs (chemicals produced by burning coal, oil, or organic material which have the ability to bind to DNA and cause mutations), and toxic heavy metals. These pollutants have been continuously entering and remaining within the Beluga Whales' ecosystem for the past several decades, leading to constant exposure with the contaminants in the water. Blubber biopsies in the 1990s showed the high levels of PCB, an organochlorine, found in free-ranging animals within the area that the whales often consume as prey, further supporting that the St. Lawrence Estuary has been a chemically stressed habitat for quite some time now (Hobbs et al., 2003). Over the course of their lifetimes, the whales accumulate enough industrial byproducts in their bodies to actually suffer long-term health problems that are often caused by the contaminants' abilities to weaken immunity and increase cancer risk (Martineau et al., 1994).

Gastrointestinal epithelial cancers, especially of the small intestine, are the most frequent type of cancer in Beluga Whales. Research found 39 malignant tumors in 35 whales with cancer being the cause of death for 20% of individuals above the age of 19 and a median diagnosis age of 48 years (Lair et al., 2015). This demonstrates the concerning frequency of cancer, as well as the late onset which often affects whales once their immunity has also been compromised, making them more susceptible to death. In other cetaceans, only 1 in 33 cancers reported or observed are gastrointestinal. Typically, Beluga Whales have similar health patterns as other cetaceans, so this statistic underscores how much of an anomaly this phenomenon and overwhelming presence actually is.

Current knowledge regarding gastrointestinal cancer in Beluga Whales points to the main cause being a contaminated environment. However, there are still gaps in understanding why exactly only this species has been impacted when other marine mammals residing in the same ecosystem remain unaffected.

To understand whether this is a unique pattern that can be solely attributed to the polluted environment, it is necessary to compare the population to other cetaceans that reside in the area, specifically, harbor porpoises and minke whales. Harbor porpoises frequent the St. Lawrence Estuary primarily during the summer and feed on small fish/invertebrates in coastal areas, similar to beluga whales, making them a good comparison species (Kingsley 2002). Minke whales are

the larger of the two and are also only seasonally present in the St Lawrence Estuary, while also sharing a similar diet to the other two.

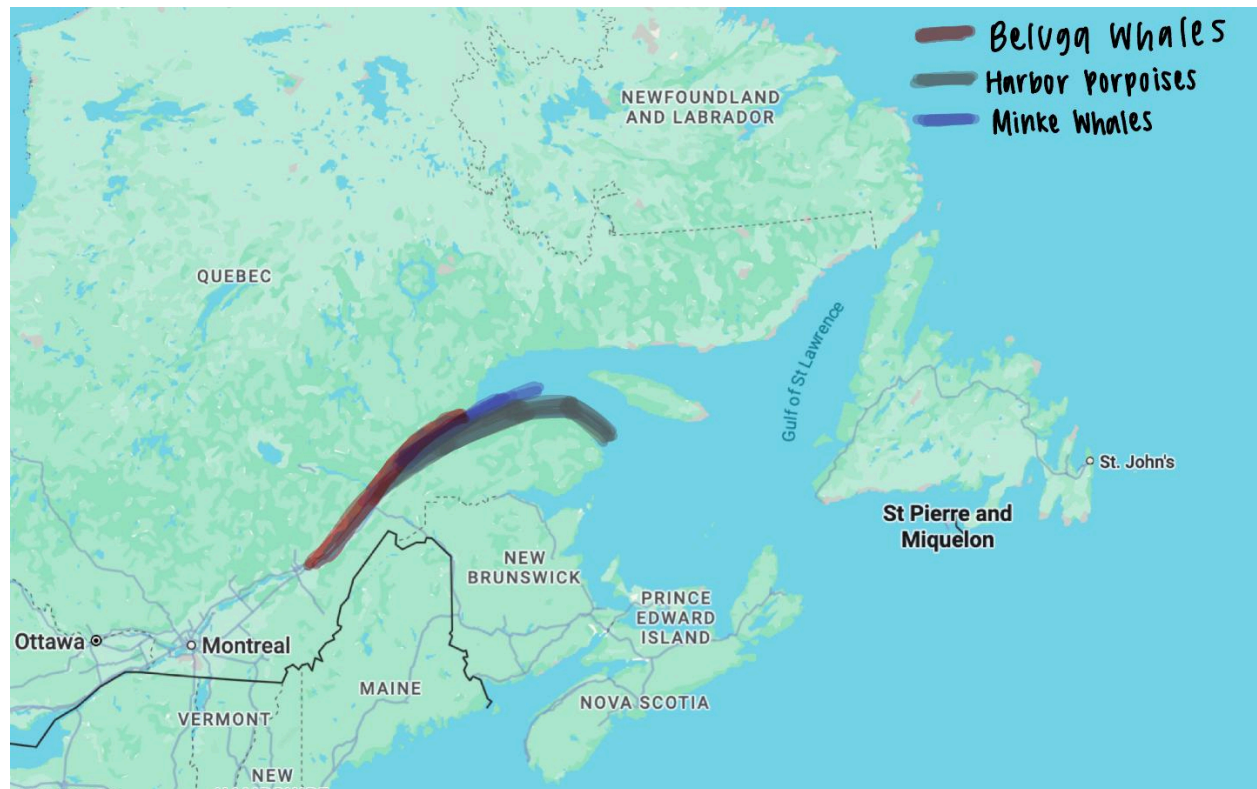


Figure 1. Summertime Habitat Distribution of Beluga Whales, Harbor Porpoises, and Minke Whales in the St. Lawrence Estuary

By comparing cancer prevalence with these cetacean populations, we can understand whether beluga whales are experiencing their unusually high cancer rates as a result of their environment, or if there is actually another undiscovered factor at play.

Objectives

Specific Question: Do the organochlorines, PAHs, and toxic heavy metals present in the St Lawrence Estuary disproportionately affect beluga whales, through their ability to cause intestinal cancer, more so than other cetaceans native to the region?

Research Synopsis: This study plans to conduct a comparative observational study to determine whether beluga whales exhibit higher intestinal cancer rates compared to other species native to the St Lawrence Estuary that are equally as exposed to the industrial contaminants causing health issues for beluga whales. The goal is to collect at least 20 post-mortem tissue samples from stranded beluga whales, harbor porpoises, and minke whales each – as many existing studies have done – across a five-year period. All tissue samples can then be analyzed for neoplastic lesions/malignancies, alongside organochlorines, PAH, and toxic heavy metal content.

Objectives:

1. Measure and compare the frequency of intestinal cancer between beluga whales, harbor porpoises, and minke whales
2. Measure and compare the concentration of organochlorines, PAH, and toxic heavy metal content in tissue samples of the three species
3. Perform a chi-squared test and logistic regression to determine whether there is a relationship between:
 - a. Exclusively beluga whales having higher pollutant concentrations and higher cancer rates generally
 - b. Exclusively beluga whales having higher pollutant concentrations and the higher intestinal cancer rates
 - c. All three species having higher pollutant concentrations and higher cancer rates generally
 - d. All three species having higher pollutant concentrations and higher intestinal cancer rates

Hypothesis: I hypothesize that beluga whales will show higher contents of the industrial pollutants and, consequently, higher malignancy correlations than harbor porpoises and minke whales. I think this is because the beluga whales' intestinal cancer rate would not have been treated as abnormally as it was if other cetaceans or nearby species exhibited similar patterns. If there were truly a correlation with the environment for this phenomenon, all species in the environment would likely be experiencing health issues, which leads me to believe this is unique to beluga whales.

Methods

The study will be a comparative, multi-species observational one to determine whether beluga whales in the St Lawrence Estuary have significantly higher intestinal cancer rates relative to other cetaceans native to the region that interact with the same industrial runoff contaminants – namely organochlorines, PAHs, and toxic heavy metals. The two other species whose tissue will be analyzed are harbor porpoises and minke whales. They allow for us to vary residence duration in the St Lawrence Estuary and diet, to see if either of these factors may be contributing to the increased cancer rates that have been historically observed by the beluga whales. Data will be collected from existing tissue archives and live sampling. This method draws inspiration from the framework used by Ylitalo et al. (2005) for a similar multi-species study regarding organochlorines and carcinoma in California sea lions.

First, we can conduct an analysis using existing post-mortem tissue samples and associated pathological reports. For the St Lawrence Estuary in particular, these can be found as part of the Quebec Marine Mammal Emergency Response Network and St Lawrence Global Observatory databases for deceased individuals that washed up between 2000 - 2024. For each observation with a confirmed neoplastic lesion or malignant tumor, we can further analyze two things: whether the individual had some form of intestinal cancer and the concentration of organochlorine, PAH, and toxic heavy metal content present in the sample. Ideally, we would study at least 20 malignant samples for each of the 3 species. This number of samples satisfies the minimum to be able to perform a chi-square test and also follows the precedent in experimental design by Ylitalo et al. (2005). Through this, we can identify whether cancer rates

are high among all regional cetaceans due to the polluted water, but intestinal cancer is just abnormally prevalent in beluga whales, as well as if there is a correlation between intestinal cancer – if present in other species as well – and the content of contaminants in the St Lawrence Estuary.

Although this longitudinal study will prove helpful for understanding contaminant data over several decades, standardized data collection is necessary for all three species simultaneously. This can be accomplished through live field sampling over three years of summer months (June, July, August) – a time when all three species are predominantly spending their time in the St Lawrence Estuary. During this period, the surrounding water can also be actively tested for loads of the three major contaminants to account for any discrepancies and changes over time. Remote biopsy techniques, in accordance with Canadian ethical standards, can be applied for collecting samples of 20 individuals per species for the three. Similar to before, after collecting these samples, evidence of neoplastic lesions or malignant tumors will be sought first. If found but the type of cancer is not confirmed, gathering a follow-up sample may be required to confirm whether it's a case of intestinal cancer. Each of the samples will also be recorded for contaminant levels as before, to identify whether there are varying amounts found by species despite the same environment. Using the historical samples, we did not have species data gathered at the same time, so contaminant levels in their surroundings may vary, but we account for that using this live sampling method. With the two combined, we are able to strengthen our ability to directly compare these species with one another.

Finally, the results from these two methods can be analyzed using a chi-squared test to determine whether intestinal cancer is significantly more common in beluga whales relative to other cetaceans – harbor porpoises and minke whales – also native to the St Lawrence Estuary. We can also use logistic regression analysis to determine whether an increased prevalence of contaminants in an individual's tissue led to increased odds for (a) cancer in beluga whales, (b) intestinal cancer in beluga whales, (c) cancer in all three species, (d) intestinal cancer in all three species. The results can also lead to future directions in terms of quantifying whether prolonged exposure impacts the impact that these contaminants have on cancer rates, as well as differences in diet. This data was collected in our study, but further analysis using other species would have to be done in order to form a more concrete conclusion regarding the correlations between those two variables and cancer prevalence in the St Lawrence Estuary.

Significance

The remarkably high intestinal cancer rates in St Lawrence Estuary beluga whales has already been observed by experts, as well as their correlation to the high volume of industrial pollutants that have contaminated the waters. The discrepancy in research, however, lies in understanding why the population is disproportionately impacted by these contaminants when compared to other cetaceans that inhabit the same, contaminated waters. The beluga whale population in the St Lawrence Estuary is already classified as endangered (Lair et al., 2005), and addressing the most common cause of death for this group is becoming increasingly important. Not only can the study help shape conservation policies for the sake of beluga whales, but findings could reveal other issues that local species are facing and urge for larger-scale ecological protection efforts.

No study yet has done a comparative analysis to see if other cetaceans that live in the St Lawrence Estuary have been facing health ramifications as severely as the cancer rates present in beluga whales in the area. The study allows for this analysis since it compares populations of beluga whales, harbor porpoises, and minke whales from the same general area – St Lawrence Estuary – which controls for the water being fairly equally contaminated for all three species. Without this direct comparison, it's difficult to tell whether the industrial pollutants present in their surroundings are uniquely contributing to the abnormally high intestinal cancer rate, or compounding with other factors that may differ from other species (ex. diet, residential patterns, genetics). Takeaways from the study can pave the way for which direction conservation policy or future research needs to take: if all three species demonstrate health implications correlated to the contaminants, general habitat remediation may be necessary. However, if not, further research can assess what additional factor is causing beluga whales to be disproportionately vulnerable to intestinal cancer and, when discerned, have relevant action taken to address that.

Apart from just beluga whale conservation, there are impacts for humans too. There are two directions from which this is possible. First, if there is some sort of relationship found between polluted waters and cancer rates in marine inhabitants, a lot of the seafood caught from this region will likely also not be healthy for human consumption. Second, given that beluga whales share a broadly similar diet to humans (primarily fish/shellfish), if the study does not find that pollution and cancer rates are linked, poor diet could still be a factor causing the intestinal issues and would require further studies into whether the prey is the culprit which could also be unhealthy for human consumption. This study and the further directions that it presents are thus very important since they have implications on human health as well.

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